

Elizabeth M Wolkovich
Department of Forest & Conservation Sciences
University of British Columbia

EDUCATION

Dartmouth College, Ph.D. 2009

Thesis: Linking community and ecosystem dynamics in invasion biology: An experimental approach in coastal sage scrub.

Advisors: Douglas T. Bolger & Kathryn L. Cottingham

Committee: Matthew P. Ayres, Ross A. Virginia, John C. Moore (outside member from NREL)

Wellesley College, B. A. 2002

Major in Biological Sciences. Minor in Russian. Study-abroad in Oaxaca, Mexico

APPOINTMENTS

Professor (CRC 2) Forest & Conservation Sciences, University of British Columbia 2026-current

Associate Professor (CRC 2) Forest & Conservation Sciences, UBC 2018-2025

Assistant Professor, Organismic & Evolutionary Biology, Harvard University 2014-2017

Biodiversity Research Centre Fellow, University of British Columbia 2012-2013

NSF Postdoctoral Fellow in Bioinformatics, University of California - San Diego 2010 - 2011

Postdoctoral Associate (6-mo), National Center for Ecological Analysis & Synthesis 2009

Lecturer, Dartmouth College, Environmental Studies Program *winter term - 2009*

SELECTED GRANTS & FELLOWSHIPS

NSERC Alliance International Collaboration (\$299,980, with Hille Ris Lambers) 2025-2028

How temporal and spatial mosaics of reproduction determine forest communities

Banff International Research Station for Mathematical Innovation and Discovery: How do trees grow in a non-stationary climate (Research in Teams) 2025

Swiss National Science Foundation (CHF 450,000, with Hille Ris Lambers) 2025-2027

MastFor: Costs and Benefits of Masting in Temperate Forest Communities

Canada Research Chair (CRC) Tier 2 in Temporal Ecology (\$500,000) 2024-2029

NSERC Discovery (\$235,000 total over 5 years) 2024-2029

How shifts in biological time and extended seasons alter plant growth across species with climate change

Canada Foundation for Innovation (\$829,000) 2019-2024

Warmer winters & future forests: How climate and phenology shape temperate woody species communities in North America and globally

Canada Research Chair (CRC) Tier 2 in Temporal Ecology (\$600,000) 2019-2024

NSERC Discovery (\$282,000 total over 6 years) 2018-2024

Predicting future forests: Understanding diverse phenological responses in North American temperate forests and globally

BC Farm Adaptation & Innovation Program (\$238,660)	2019-2023
Modeling winegrape phenology for a warming Okanagan	
BRC Grant for Catalyzing Research Clusters (UBC, \$25,360)	2023
Accelerating Science through Generative Bayesian modeling (working group & training)	
Faculty of Forestry Seed Money Grant (UBC, \$19,880)	2023
Warmer springs and stagnate growth (Proof of concept funding)	
Milton Fund (Harvard University, \$39,766)	2015
Phenological responses to climate change: Plasticity, local adaptation and the diversity of species' responses	
Radcliffe Institute Exploratory Seminar (\$20,930, Ailene Ettinger co-PI)	2015-2016
Predicting future springs: Reconciling experimental and observational approaches for climate change impacts	
Biodiversity Postdoctoral Fellowship (UBC, \$100,000)	2012 - 2013
NCEAS Working Group (co-PI with Benjamin I. Cook, \$70,250 plus extension)	2010 - 2012
Forecasting phenology: Integrating ecology, climatology, and phylogeny to understand plant responses to climate change	
NSF Postdoctoral Research Fellowship in Biology (Bioinformatics) , \$123,000)	2010 - 2011
Phenology of plant invasions: How changing seasons and temporal niches assemble plant communities	
EPA Science to Achieve Results (STAR) Fellowship (\$111,000)	2006 - 2009
Impact of invasive plants on detrital food webs	
Exotic/Invasive Pests and Diseases Research (UC-IPM, \$23,990)	2006 - 2008
Title: Impact of exotic Mediterranean annual grasses on detrital food webs	
GAANN Fellowship, Dartmouth College (\$30,000)	2005 - 2006
Center for Invasive Plant Management Seed Money Grant (\$4,939)	2005 - 2006

SELECTED AWARDS & HONORS

Killam & ETH Rubel Foundation Fellowships	2022
Early Career Fellow , Ecological Society of America (ESA)	2016-2020
Tansley Medal Runner-up , <i>New Phytologist</i>	2013
Murray F. Buell Award for best student oral presentation (ESA)	2009
Graduate Student Filene Teaching Award , Dartmouth College	2009

SELECTED PUBLICATIONS

*Denotes lab member; note that last author is senior author.

Baumgarten,* F., Aitken, S., Vitasse, Y. Guy, R. D. & **E. M. Wolkovich**.2026. (In)eterminacy in woody plants: limits and opportunities for timing growth in a changing climate. *Ecology Letters*. 29 (6) e70435

E. M. Wolkovich, Davies, T.J., Pearse, W. & M. Betancourt. 2026. A four-step simulated-based workflow for improving ecological science. *Philosophical Transactions of the Royal Society A*. 384 (2321): 20250252

- Davies, T.J., Hille Ris Lambers, J. & **E. M. Wolkovich**. 2025. The Janzen-Connell hypothesis and seed masting. *Trends in Ecology & Evolution*. 40 (12): 1190-1198
- Wolkovich, E. M.**, Ettinger*, A.K., Chin, A., Chamberlain*, C. J., Baumgarten*, F., Pradhan, K., Manzanedo, R.D. & J. Hille Ris Lambers. 2025. Why longer seasons with climate change may not increase tree growth. *Nature Climate Change*. 15: 1283–1292
- Wolkovich, E. M.**, Rouleau-Desrochers*, C. García de Cortázar-Atauri, I., Walker, M. A. & T. Lacombe. Building a more predictive model of terroir for the Anthropocene. *Plants, People & Planet*: doi.org/10.1002/ppp3.70055
- Van der Meersch*, V. and **E. M. Wolkovich**. 2025. Summer solstice optimizes the thermal growing season. *PNAS*. 122(23):e2506796122.
- Wolkovich, E. M.**, Cook, B.I., García de Cortázar-Atauri, I., V. Van der Meersch,* Lacombe, T., Marcal, C. & I. Morales-Castilla.* 2025. Uneven impacts of climate change around the world and across the annual cycle of winegrapes. *PLOS Climate*. 4(5): e0000539. doi.org/10.1371/journal.pclm.0000539.
- Pearse, W., Davies, T.J. & **E. M. Wolkovich**. 2025. How to define, use, and interpret Pagel's λ (lambda) in ecology and evolution. *Global Ecology & Biogeography*. 34: e70012. https://doi.org/10.1111/geb.70012
- Wolkovich, E. M.** The problem of terroir in the Anthropocene. 2025. *Harvard Data Science Review*. 7(2). https://doi.org/10.1162/99608f92.d605e50f
- Cleland[§], E.E. & **E. M. Wolkovich**[§]. 2024. Effects of Phenology on plant community assembly and structure. *Annual Review of Ecology, Evolution & Systematics*. 55(1), 471-492. [§]Both authors contributed equally.
- Morales-Castilla*, I., Davies, T. J., Legault*, G., Buonaiuto*, D. M., Chamberlain*, C.J., Ettinger*, A.K., Garner*, M., Jones*, F. A., Loughnan*, D. L., Pearse, W. Sodhi*, D. & **E. M. Wolkovich**. 2024. Phylogenetic estimates of species-level phenology improve ecological forecasting. *Nature Climate Change*. 14(9): 989-995.
- Zeng*, Z.A. and **Wolkovich, E.M.** 2024, Weak evidence of provenance effects in spring phenology across Europe and North America. *New Phytologist*. 242(5): 1957-1964
- Jones*, F. J. , Bogdanoff, C. & **E. M. Wolkovich**. 2024. The role of genotypic and climatic variation at the range edge: A case study in winegrapes. *American Journal of Botany* (111): e16270
- Buonaiuto*, D. M. & **E. M. Wolkovich**. 2023. Contrasting responses to climate variability generate seasonal priority effects between native and invasive forest herbs. *Journal of Ecology* (111): 1711-1721.
- Wolkovich, E. M.**, Chamberlain*, C.J., Buonaiuto*, D. M., Ettinger*, A.K. & I. Morales-Castilla.* 2022 Integrating experiments to predict interactive cue effects on spring phenology with warming. *New Phytologist*. 235: 1719-1728.
- Wolkovich, E. M.**, Auerbach, J. A., Chamberlain*, C.J., Buonaiuto*, D. M., Ettinger*, A.K., Morales-Castilla*, I. & A. Gelman. 2021. A simple explanation for declining temperature sensitivity with warming. *Global Change Biology*: 10.1111/gcb.15746.

- Wolkovich, E. M.** & Donahue, M. 2021. How phenological tracking shapes species and communities in non-stationary environments. *Biological Reviews* (96) 2810-2827.
- Eyster, H.* & **E. M. Wolkovich**. 2021. Comparisons in the native and introduced ranges reveal little evidence of climatic adaptation in germination traits. *Climate Change Ecology*: doi.org/10.1016/j.ecochg.2021.100023
- Chamberlain*, C.J & **E. M. Wolkovich**. 2021. Late spring freezes coupled with warming winters alter temperate tree phenology and growth. *New Phytologist*: (231) 987–995.
- Ettinger*, A.K., Buonaiuto*, D. M., Chamberlain*, C.J., Morales-Castilla* & **E. M. Wolkovich**. 2021. Spatial and temporal shifts in photoperiod with climate change. *New Phytologist*: 230 (2), 462–474.
- Ettinger*, A.K., Chamberlain*, C.J., Buonaiuto*, D. M., Morales-Castilla*, I., Flynn*, D.F.B., Savaas*, T., Samaha*, J. & **E. M. Wolkovich**. 2020. Winter temperatures predominate in spring phenological responses to warming. *Nature Climate Change*: 10 (12), 1137–1142.
- Chamberlain*, C.J., Cook, B.I., Morales-Castilla*, I. & **E. M. Wolkovich**. 2020. Climate change reshapes the drivers of false spring risk across European trees. *New Phytologist*: 229 (1), 323–334
- Merrill*, N.K., I., García de Cortazar-Atuari, I., T., Parker, A., Walker, M. A. & **E. M. Wolkovich**. 2020. Exploring grapevine phenology and high temperatures response under controlled conditions. *Frontiers in Environmental Science* (8) 220: 10.3389/fenvs.2020.516527
- Morales-Castilla*, I., García de Cortazar-Atuari, I., Cook, B. I., Lacombe, T., Parker, A. van Leeuwen, C. & **E. M. Wolkovich**. 2020. Cultivar diversity buffers winegrowing regions from climate change losses. *PNAS*: 117 (6), 2864–2869.
- Chamberlain*, C.J., Cook, B.I., García de Cortazar-Atuari, I. & **E.M. Wolkovich**. 2019. Rethinking false spring risk. *Global Change Biology* 25(7): 2209–2220.
- Joly, S, D. F. B. Flynn* & **E. M. Wolkovich**. 2019. On the importance of accounting for intraspecific genetic correlations in multi-species studies. *Methods in Ecology & Evolution* 10(7): 994–1001.
- Davies, T. J., J. Regetz, **E. M. Wolkovich** & B. J. McGill. 2019. Phylogenetically weighted regression: a method for modeling non-stationarity on evolutionary trees. *Global Ecology & Biogeography* (28) 2: 275–285.
- Ettinger*, A. K., Chuine I., Cook, B. I., Dukes, J. S., Ellison, A. M., Johnston, M. R., Panetta, A. M., C.R. Rollinson, C.R., Vitasse, Y., & **E.M. Wolkovich**. 2019. How do climate change experiments actually change plot-scale climate? *Ecology Letters* 22(4): 748–763.
- Flynn*, D. F. B. & **E.M. Wolkovich**. 2018. Temperature and photoperiod drive spring phenology across all species in a temperate forest community. *New Phytologist* (219) 1353–1362.
- Wolkovich, E. M.**, I. García de Cortázar-Atauri, Morales-Castilla, I., K. A. Nicholas & T. Lacombe. 2018. From Pinot to Xinomavro in the world's future winegrowing regions. *Nature Climate Change*: 8:29–37.
- Wolkovich, E. M.**, D. O. Burge, M. A. Walker & K. A. Nicholas. 2017. Phenological diversity provides opportunities for climate change adaptation in winegrapes. *Journal of Ecology* 105:905–912.
- Cook, B. I. & **E. M. Wolkovich**. 2016. Climate change decouples drought from early winegrape

harvests in France. *Nature Climate Change* 6:715-719.

Wolkovich, E. M., Cook, B. I., McLauchlan, K. K. & T. J. Davies. 2014. Temporal ecology in the Anthropocene. *Ecology Letters* 17(11): 1365–1379.

Wolkovich, E. M., B. I. Cook & T. J. Davies. 2014. Progress towards an interdisciplinary science of plant phenology: Building predictions across space, time and species diversity. *New Phytologist* 201: 1156–1162.

Wolkovich, E. M. & E. E. Cleland. 2014. Phenological niches and the future of invaded ecosystems with climate change. *AoB Plants* doi:10.1093/aobpla/plu013.

Wolkovich, E. M. & 18 co-authors. 2012. Warming experiments underpredict plant phenological responses to climate change. *Nature* 485 (7399): 494-497. (Recommended by Faculty of 1000)

Wolkovich, E. M., Regetz, J. & M. I. O'Connor. 2012. Advances in global change research require open science by individual researchers. *Global Change Biology* 18(7): 2102-2110.

Cook, B. I., **Wolkovich, E. M.** & C. Parmesan. 2012. Divergent responses to spring and winter warming explain community level flowering trends. *Proceedings of the National Academy of Sciences* 109(3): 9000-9005.

Cleland, E. E., J. M. Allen, T. M. Crimmins, J. A. Dunne, S. Pau, S. E. Travers, E. S. Zavaleta & **E. M. Wolkovich**. 2012. Phenological tracking enables positive species responses to climate change. *Ecology* 93(8): 1765-1771.

Davies, T.J., Kraft, N.B.J., Salamin, N. & **E. M. Wolkovich**. 2012. Incompletely resolved phylogenetic trees inflate estimates of phylogenetic conservatism. *Ecology* 93(2): 242-247.

Wolkovich, E. M. & E. E. Cleland. 2011. The phenology of plant invasions: A community ecology perspective. *Frontiers in Ecology & the Environment* 9(5): 287-294. (Recommended by Faculty of 1000)

CURRENT TEACHING

Professor , University of British Columbia	<i>2018-onward</i>
Ecology in a changing climate (CONS 310)	
Experimental design and hierarchical model building with Bayesian inference (FRST 507C-202)	
Bayesian modeling group at UBC (meets 2X/monthly, 2020-2021, 2025)	

SELECTED UNIVERSITY & OTHER PROFESSIONAL SERVICE

Guest editor for <i>PNAS</i>	<i>2025</i>
Special Issue Editor for <i>Philosophical Transactions A</i> 'Workflow for Applied Data Analysis' with co-organizers and editors: Andrew Gelman, Ali Vehtari, Richard McElreath	<i>2025</i>
NCEAS Environmental Data Science Advisory Board	<i>2020 - 2026</i>
<i>Climate Change Ecology</i> Science Advisory Board	<i>2021- onward</i>
Dryad Scientific Advisory Committee	<i>2020-2022</i>
Canadian Institute for Ecology & Evolution management board member	<i>2019 - 2025</i>
<i>Climate Change Ecology</i> Special Issue Editor	<i>2020-2021</i>
Stan (probabilistic programming language) non-profit board member	<i>2016 - 2019</i>

RECENT SELECTED PRESENTATIONS

Wolkovich, E. M. 2025. Recovery from climate-related events: Matching variety to climate. Canadian Grapevine Certification Network Webinar Series (Virtual).

Wolkovich, E.M. 2024. How winter temperatures may control global forecasts of forest carbon sequestration. AGU Fall Meeting (USA).

Wolkovich, E.M. 2024. Lessons on variety research with climate change. BC Horticultural Show (Canada).

Wolkovich, E. M. 2022. The Problem of Time in Climate Change Ecology at University of Alcalá.

Wolkovich, E. M. 2022. Modelling grapevine phenology and impacts of climate change on viticulture International Cool Climate Wine Symposium (Keynote for viticulture),

Wolkovich, E. M. 2022. Modeling grapevine phenology & Impacts of climate change on viticulture. International Cool Climate Wine Symposium at Brock University.

SELECTED PUBLISHED DATA

*All lab data are public and we work to make all data we use publicly available, following our lab Data Management Plan, which itself has been widely re-used. All data on Knowledge Network for Biocomplexity.

Buonaiuto, D. & E. M. Wolkovich. 2021. Environmental drivers of flower-leaf sequences variation in temperate woody plants. (doi:10.5063/PG1Q4B)

Karouba, H & E. M. Wolkovich. 2020. Disconnects between ecological theory and data in phenological mismatch research. Dryad (doi.org/10.5061/dryad.7pvmcudpz)

Chamberlain, C. & E. M. Wolkovich. Phenology and growth data across eight temperate tree and shrub species exposed to late spring freezes coupled with warming winters. (doi:10.5063/F47MJ4)

Chamberlain, C. 2020. False spring risk of six European temperate trees from 1950-2016. (doi:10.5063/JW8C90)

Merrill, N. & E. M. Wolkovich. 2019. Budburst and leafout phenology of 50 varieties of winegrapes (*Vitis vinifera* subsp. *vinifera*) in greenhouse with heat tolerance of flowering study. (doi:10.5063/F1TM78HS)

Ignacio Morales-Castilla, Marta Fernández-Pastor, & Wolkovich, E.M.. 2019. Winegrowing Regions of the World 2007. (doi:10.5063/F1SX6BH1)

SELECTED OUTREACH

International Cherry Blossom Prediction Competition 2022 - onward

Open competition to predict date of cherry blossoms in four cities across the globe based on all public cherry blossom data, developed with J. Auerbach & D. Kepplinger (Statistics, George Mason University)

Tree Spotters citizen science group - USA National Phenology Network 2016 - onward

Developed and led new group for the USA National Phenology Network on widespread woody tree species and focused at the Arnold Arboretum; I continually give lectures and meet with members. Awarded USA-NPN's 2021 PhenoChampion.

Winegrape growers' groups

Give 2-6 lectures as requested each year (some listed above)